BIG IDEAS MATH • GRADE 6 • CHAPTER 2

Fractions & Decimals: Dividing Fractions & Reciprocals

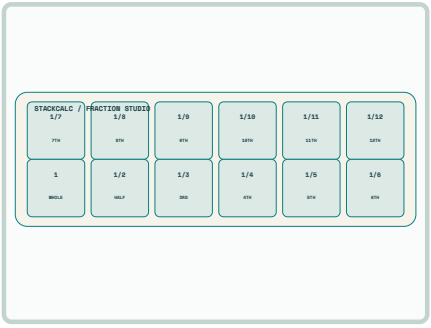
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.6.NS.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Fraction Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place one 3/4 block (63mm) and 1/8 blocks (10.5mm). Count how many 1/8 blocks fit into 3/4 (6 blocks).

Sketch Model on Paper:
Draw a 3/4 bar partitioned into eighths: 6 equal eighths.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	FDISP	Fraction Studio	
1.	3/4	X: 3/4	Key dividend 3/4 (.3
2.	ENTER	Y: 3/4 | X: 3/4	Stack lift
3.	1/8	Y: 3/4 | X: 1/8	Key divisor 1/8 (.1.
4.	÷	X: 6	Divide: 3/4 ÷ 1/8 =

EXPECTED LCD X: 6

1 **Divide 1/2 by 1/4 using Fraction Studio.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Divide 2/3 by 1/6 using Fraction Studio.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Divide 3/5 by 1/10 using Fraction Studio.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Evaluate 4/5 ÷ 2/5 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 2

Fractions & Decimals: Dividing Fractions & Reciprocals • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Fraction Studio Proportional Towers

CAD Blueprint: fraction_tray

Target Device: SC-6 / SC-32 Bridge

CAD / STL Source File:

Education/scad/fraction_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal 3x1 Bin & Color-Coded Towers (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Divide 1/2 by 1/4 using Fraction Studio.** Standard: CCSS.MATH.6.NS.A.1	2	[. 1 . 2] [ENTER] [. 1 . 4] [÷]
2	**Divide 2/3 by 1/6 using Fraction Studio.** Standard: CCSS.MATH.6.NS.A.1	4	[. 2 . 3] [ENTER] [. 1 . 6] [÷]
3	**Divide 3/5 by 1/10 using Fraction Studio.** Standard: CCSS.MATH.6.NS.A.1	6	[. 3 . 5] [ENTER] [. 1 . 10] [÷]
4	**Evaluate 4/5 ÷ 2/5 on the stack.** Standard: CCSS.MATH.6.NS.A.1	2	[. 4 . 5] [ENTER] [. 2 . 5] [÷]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Students often invert the dividend instead of the divisor.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.